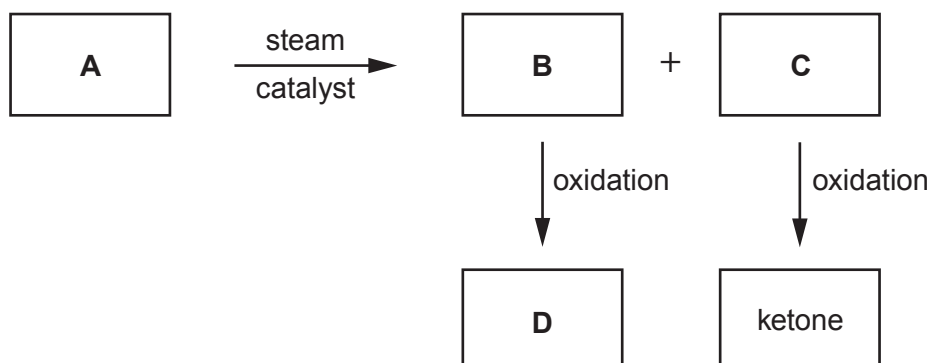


9. Look at the reaction scheme below and other information about compounds **A** to **D**.



A is a straight chain hydrocarbon. It contains 85.7 % carbon by mass.

B and **C** are structural isomers.

D is soluble in water. 0.93 g of **D** is dissolved in water to make 100 cm³ of solution. 25.0 cm³ of this solution is titrated against 0.100 mol dm⁻³ aqueous sodium hydroxide. 26.40 cm³ of sodium hydroxide is needed for neutralisation. **D** reacts with sodium hydroxide in a ratio of 1:1.

- (a) Calculate the empirical formula of **A**.

[2]

Empirical formula

